

MADLEY, United KingdomWMO: **035210**Lat: **52.033N**Lon: **2.850W**Elev: **78**StdP: **100.39**Time Zone: **0.00 (EUW)**Period: **86-97**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.3	-2.8	-5.6	2.4	-3.1	-4.2	2.7	-1.3	13.9	6.7	12.3	9.5	0.5	60	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	25.2	18.8	23.5	18.0	21.9	17.1	19.7	23.7	18.6	22.6	17.6	21.1	2.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.2	21.7	17.1	12.3	20.5	16.1	11.5	19.4	56.4	23.7	52.8	22.7	49.7	21.2	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.4	8.8	7.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			-8.1	21.1	2.8	1.0	-10.1	21.9	-11.7	22.5	-13.2	23.0	-15.2	23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	5.2	4.5	6.6	8.1	11.2	14.2	16.3	15.5	12.7	10.2	7.0	5.5
	DBStd	5.04	3.35	3.80	2.87	2.84	2.76	2.75	2.10	2.67	2.42	3.13	3.48	4.04
	HDD10.0	793	149	154	109	71	19	2	0	0	5	37	102	147
	HDD18.3	3136	406	386	363	307	223	128	69	94	169	252	340	399
	CDD10.0	716	1	1	4	14	55	126	194	172	87	43	11	7
	CDD18.3	16	0	0	0	0	0	3	6	7	0	0	0	0
	CDH23.3	161	0	0	0	0	3	39	58	59	2	0	0	0
	CDH26.7	19	0	0	0	0	0	4	3	12	0	0	0	0
(14) Wind	WSAvg	2.7	3.6	3.7	3.5	2.8	2.3	2.0	2.1	1.9	2.0	2.3	2.5	3.2
(15) Precipitation	PrecAvg	808	89	66	52	56	60	54	55	59	60	88	82	86
	PrecMax	1049	183	174	97	158	107	148	165	149	148	174	158	167
	PrecMin	528	8	4	10	6	6	9	7	9	11	14	39	24
	PrecStd	126	38	41	26	38	24	35	36	36	35	41	31	40
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	12.9	15.3	19.5	23.7	26.9	27.1	27.4	22.7	18.8	15.7	14.5
		MCWB	11.0	10.7	11.6	14.7	17.0	19.2	18.7	19.2	17.3	15.7	13.7	12.3
	2%	DB	11.8	11.9	13.2	16.6	21.5	24.0	25.2	24.7	20.0	16.9	14.6	13.6
		MCWB	9.8	9.9	10.5	12.6	16.1	18.4	18.7	19.0	15.9	14.6	12.8	11.7
	5%	DB	11.0	10.9	12.0	14.7	19.1	22.1	23.5	22.6	18.7	15.7	13.1	12.7
		MCWB	9.4	9.1	9.8	11.5	14.5	17.0	18.1	18.0	15.0	13.8	11.7	11.0
	10%	DB	10.0	9.9	11.1	13.3	16.8	20.0	21.9	20.7	17.4	14.8	12.2	11.4
		MCWB	8.6	8.3	9.0	10.5	13.1	15.9	17.1	16.7	14.2	13.2	10.8	9.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	11.3	12.2	15.2	17.8	20.3	20.8	21.2	18.1	16.3	14.3	12.7
		MCDB	12.4	12.5	14.4	19.1	22.6	24.4	25.7	25.1	21.1	17.9	15.5	14.0
	2%	WB	10.4	10.2	11.0	13.0	16.3	18.7	19.5	19.7	16.8	15.1	13.1	12.0
		MCDB	11.5	11.4	12.8	15.4	20.9	23.3	23.6	23.6	19.3	16.3	14.2	13.3
	5%	WB	9.6	9.3	10.2	11.9	14.9	17.3	18.4	18.5	15.7	14.3	12.1	11.1
		MCDB	10.8	10.7	11.8	14.6	18.4	21.5	22.4	22.1	17.7	15.5	13.0	12.6
	10%	WB	8.8	8.4	9.4	10.8	13.7	16.1	17.5	17.3	14.8	13.5	11.0	10.0
		MCDB	10.0	9.8	11.0	13.0	16.3	19.7	21.2	20.1	16.9	14.7	12.0	11.3
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.3	6.3	7.7	8.7	8.6	8.6	8.5	8.2	6.5	5.7	4.6
		MCDBR	6.0	6.1	7.5	10.9	13.2	13.1	12.9	11.7	10.8	7.6	5.5	5.3
		MCWBR	5.6	5.3	5.6	7.2	8.3	7.6	7.2	6.4	7.1	5.8	4.9	4.8
	5% WB	MCDBR	5.8	5.7	7.0	10.2	12.1	11.8	11.1	10.3	8.7	6.6	5.5	5.0
		MCWBR	5.7	5.2	5.7	6.9	7.9	7.5	6.9	6.4	6.5	5.4	5.2	4.6
(40) Clear-Sky Solar Irradiance	taub		0.322	0.338	0.371	0.394	0.400	0.393	0.390	0.375	0.365	0.349	0.323	0.312
	taud		2.339	2.368	2.277	2.227	2.248	2.300	2.311	2.370	2.399	2.415	2.392	2.316
	Ebn at Noon		670	775	814	839	852	861	856	851	816	751	667	611
	Edh at Noon		63	81	106	125	129	124	120	108	93	76	59	54
(44) All-Sky Solar Radiation	RadAvg		0.75	1.42	2.54	3.91	4.83	5.20	4.86	3.99	3.02	1.70	0.90	0.58
	RadStd		0.07	0.16	0.31	0.45	0.27	0.56	0.50	0.33	0.23	0.15	0.09	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (79 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page